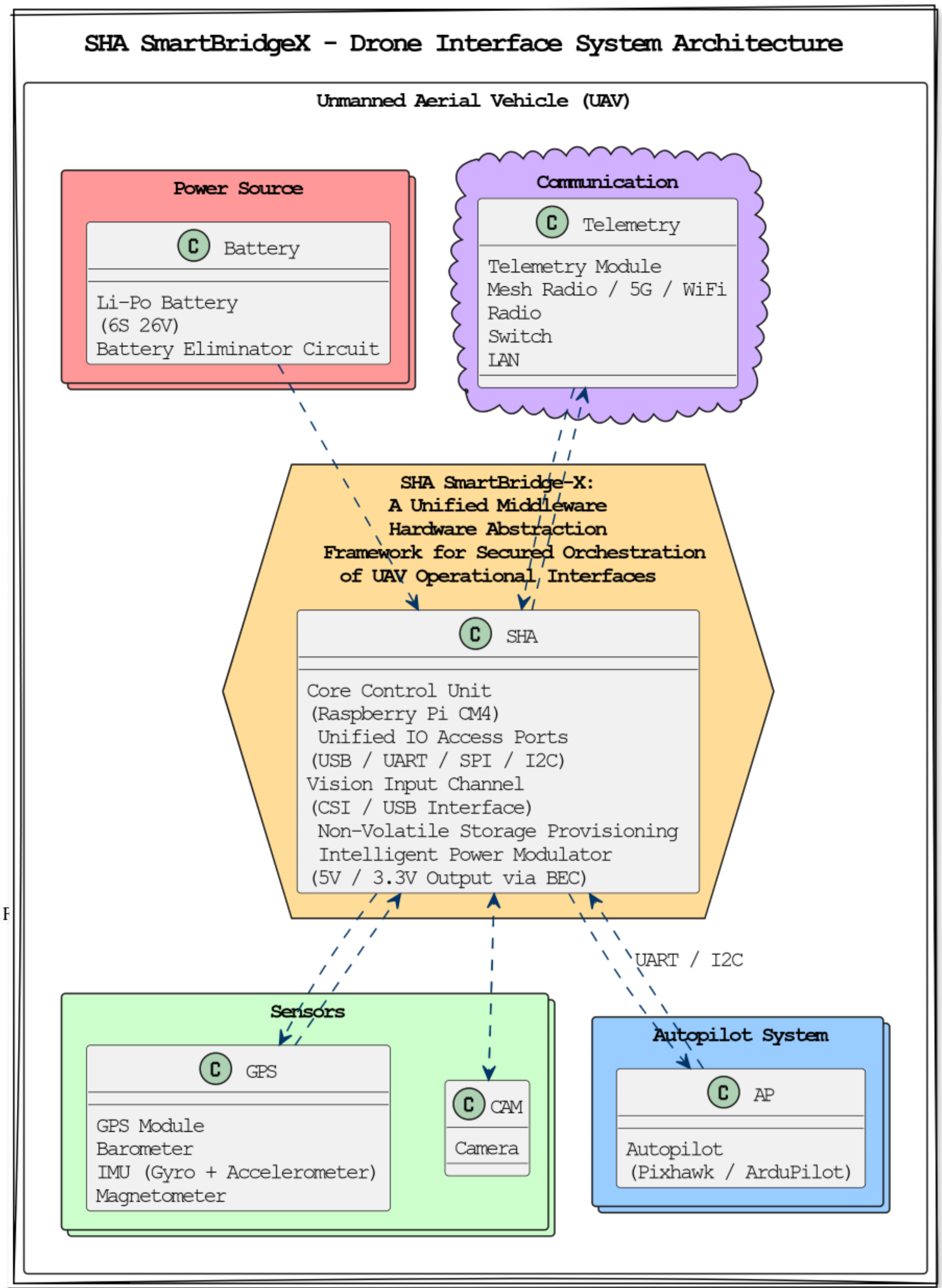
SHA SmartBridge-X Proposal

Title:

"SHA SmartBridge-X: A Unified Middleware Hardware Framework for Secure Orchestration of UAV Functional Interfaces"



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PICO Framework Compliance:

Problem (P): Modern UAV systems require multiple independent modules—autopilot, telemetry, sensors, storage, and power regulation—resulting in integration complexity, unreliability, and inefficiency.

Intervention (I): SHA SmartBridge-X provides a unified, middleware hardware framework to securely manage and orchestrate UAV functional interfaces with built-in tamper resistance and access control features.

Comparison (C): Unlike current solutions that rely on loosely connected breakout modules, this product delivers an indigenous, scalable orchestration layer that improves reliability and simplifies development cycles.

Outcome (O): Enhanced integration reliability, reduced deployment time, and foundational security for UAVs used in academic, defense, and commercial sectors.

Novelty & Why This Product:

SHA SmartBridge-X is a securely engineered hardware framework designed to orchestrate diverse functional interfaces within UAV ecosystems enabling streamlined interaction across control logic, telemetry, environmental awareness, and data handling layers. It integrates indigenous design practices with foundational security features such as tamper detection and interface-level safeguards. The solution offers a unified orchestration layer for real-time UAV operations, moving beyond traditional modular integration approaches.

Target Audience / Demand Justification:

SHA SmartBridge-X is designed for startups, research labs, defense R&D, and UAV manufacturers requiring robust and reliable integration solutions.
Its secure and streamlined design supports high-demand applications such as BVLOS operations, agri-drones, and AI-powered surveillance systems.
This innovation reduces time-to-deployment significantly, enabling broader adoption and scalability across India's growing drone ecosystem.